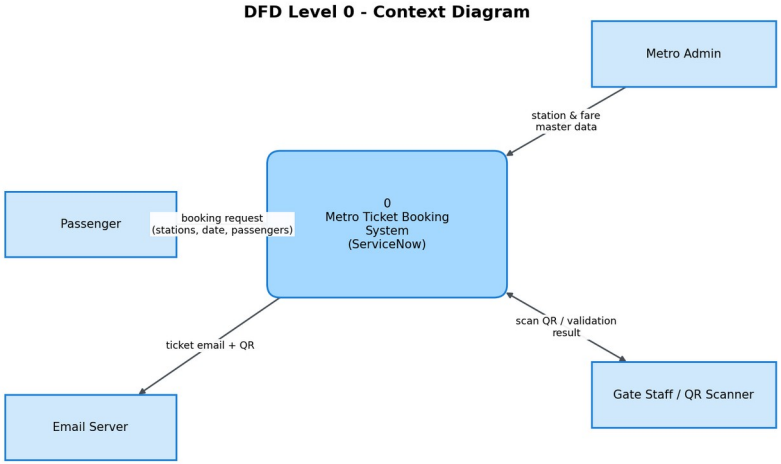
Project Design Phase-II
Data Flow Diagram & User Stories

Date	15/10/2026
Team ID	
Project Name	Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

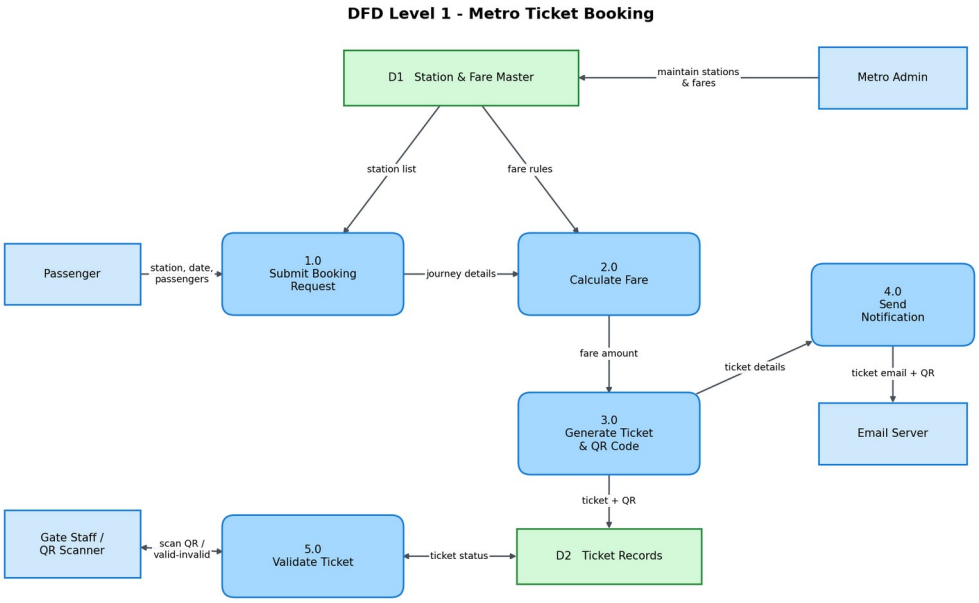
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: DFD Level 0 (Context diagram)



DFD Level 1



Flow description:

- The passenger submits station, date and passenger details through the Service Catalog (Process 1.0).
- Station list and fare rules are read from the Station & Fare Master (D1), which the administrator maintains.
- Process 2.0 calculates the fare; Process 3.0 creates the ticket, generates the QR code and stores it in Ticket Records (D2).
- Process 4.0 sends the ticket email with the QR code through the email server.
- At the gate, Process 5.0 checks the scanned QR code against Ticket Records and returns valid or invalid.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my name, email and password.	I can register and access my account / dashboard	High	Sprint-1
	Login	USN-2	As a user, I can log into the application by entering my email and password.	I can log in and see the Service Portal home page	High	Sprint-1
	Book Ticket	USN-3	As a user, I can select the source and destination stations from a list.	Both stations are mandatory and cannot be the same	High	Sprint-1
		USN-4	As a user, I can choose the journey date and number of passengers.	Past dates are not accepted; passenger count is 1 or more	High	Sprint-1
	Fare	USN-5	As a user, I can see the fare calculated automatically before I confirm.	Fare equals the fare-matrix amount multiplied by passengers	High	Sprint-1
	Digital Ticket	USN-6	As a user, I receive a digital ticket with a unique QR code immediately after booking.	Ticket number and QR code are generated and stored on the ticket record	High	Sprint-2
		USN-7	As a user, I can view and download my tickets from the My Tickets page.	All my tickets are listed with status	Medium	Sprint-2
	Notifications	USN-8	As a user, I receive a confirmation email with ticket details and QR code.	Email is received within a minute of booking	High	Sprint-2
		USN-9	As a user, I receive a reminder before my journey date.	Reminder is sent one day before the journey	Low	Sprint-3
	Ticket Management	USN-10	As a user, I can cancel an unused ticket.	Ticket status changes to Cancelled and a cancellation email is sent	Medium	Sprint-3
Customer (Web user)	Book Ticket	USN-11	As a web user, I can book a ticket from the portal using a laptop or desktop browser.	All booking steps work on the web with the same result as mobile	High	Sprint-1
Metro Gate Staff	Ticket Validation	USN-12	As gate staff, I can scan the QR code to check that a ticket is valid.	Valid ticket is accepted once; used or cancelled ticket is rejected	High	Sprint-2
Customer Care Executive	Support	USN-13	As a customer care executive, I can search a ticket by number and resend it to the passenger.	Ticket details are shown and the email is re-sent	Medium	Sprint-3
Administrator	Master Data	USN-14	As an administrator, I can add and edit stations.	New station appears in the booking form	High	Sprint-1
		USN-15	As an administrator, I can maintain the fare	Updated fare is used for new bookings	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			between stations.			
	Dashboard	USN-16	As an administrator, I can view bookings and revenue reports.	Dashboard shows tickets booked per day and per route	Medium	Sprint-3